Heat-Aware Walking Routes under Distance Limits: A Washington, DC Pilot

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Abstract

Heat-aware walking routes trade modeled thermal exposure against extra distance. We compare four environmental objectives with shortest distance on a Washington, DC network, using Universal Thermal Climate Index (UTCI) and shadow rasters at one modeled hour. Five methods, 500 origin–destination pairs, and four distance allowances give 10,000 attempts at 5 m sampling. Mean reductions in length-weighted UTCI exceedance above 26°C are 0.91–0.94 K, with 4.59–6.34% extra distance. Switching from 20 m to 5 m changes 70–76% of environmental paths and lowers mean exceedance by 0.06–0.10 K when both sets are scored at 5 m. We report shortest-path fallback, coefficient and threshold checks, and search losses against exact references. The contribution is a reproducible comparison of objective choices within one capped solver. Results describe one modeled field and remain unvalidated against pedestrian measurements.

Keywords

pedestrian routing, UTCI, shade, detour constraints, computational benchmark

1 Introduction

Two walking routes of similar length can have different shade and thermal conditions. A route that avoids exposed streets may reduce modeled heat exposure, but a long detour also adds walking. An explicit distance allowance makes this tradeoff understandable: a traveler can ask for a cooler route no more than 10% longer than the shortest path.

This paper examines that formulation in a Washington, DC pilot. It attaches modeled UTCI and direct-shadow rasters to a pedestrian graph, then compares objectives that emphasize UTCI, shade, or a composite of environmental features. We study how these objective settings behave within one capped search implementation, using established hard-detour constraints and cost–distance labels.

We compare five methods on the same 500 origin–destination (OD) pairs and four distance allowances, retaining shortest-path fallback in the results. The outcomes are modeled exceedance, added distance, and shade. We report both average exceedance and its length-scaled total, because a cooler average can accompany more total walking through heat. Coefficient checks, denser rerouting, and exact-reference tests examine the numerical choices behind that comparison.

2 Related Work

Shade-aware routing predates this study. Vartholomaios derives least-cost paths from SOLWEIG shadows [23]; Da et al. use image-derived shade [5]; and Kolaxidis et al. incorporate solar exposure into pedestrian routing [16]. CoolWalks studies sunlit-distance penalties and the difference between centerline and detailed pedestrian networks [25]. Yang et al. incorporate time-varying building shadows [26], whereas this pilot holds the environmental field fixed.

Other work uses thermal indices. Colaninno et al. combine sidewalk mobility and microclimate modeling [4]. Patel et al. rank Google Maps candidate routes using SOLWEIG-GPU UTCI [22]. CoolPaths minimizes summed edge length times Physiological Equivalent Temperature (PET) under a detour limit [24]. Here, the primary reported metric is length-normalized UTCI exceedance. The routing formulation follows established resource-constrained shortest-path methods [14].

UTCI converts meteorological and radiative conditions into an equivalent temperature under standardized reference assumptions [3]. SOLWEIG models urban radiation and mean radiant temperature [18]. Evidence that heat influences route choice [1] and that pedestrians seek shade [17] motivates these routing preferences.

3 Routing Pipeline

3.1 Graph and distance budget

Let G_0 be the original directed walking multigraph. For a route P, let $L(P) = \sum_{e \in P} l_e$, where l_e is edge length in metres. For an OD pair, define the original-network shortest distance L_0^* and the final ceiling

$$L_0^* = \min_{P \subset G_0} L(P), \qquad D_0 = (1 + \delta) L_0^*. \tag{1}$$

The minimization is over directed paths connecting that pair. The allowance $\delta \in \{0.05, 0.10, 0.20, 0.30\}$ limits proportional extra distance.

To reduce feature-extraction work, a node v is retained when

$$d_{G_0}(o, v) + d_{G_0}(v, d) \le D_0 + 50\,\text{m}, \tag{2}$$

where d_G denotes shortest-path distance on graph G. The pipeline then assigns raster-backed features and removes edges lacking valid UTCI samples. The resulting feature graph is G_f. Network processing uses OpenStreetMap, OSMnx, and NetworkX [2, 13, 21].

3.2 Sampling and reported exposure

Each source segment in an edge geometry is subdivided to a maximum spacing of 5 m, retaining endpoints. Coordinates are transformed into each raster's coordinate reference system before sampling. Non-finite and NoData samples are omitted. Edges without UTCI samples are removed; edges without direct-shadow samples use the surrogate in Appendix B.

For UTCI samples u_{ei} on edge e, define

$$\bar{u}_e = \frac{1}{n_e^u} \sum_i u_{ei}, \qquad u_e^{\max} = \max_i u_{ei}, \tag{3}$$

$$h_e = \frac{1}{n_e^u} \sum_i \max(0, u_{ei} - \tau), \quad \tau = 26^\circ\mathrm{C}. \tag{4}$$

For valid shadow samples b_{ej}, the direct-shade fraction is $s_e^{\mathrm{dir}} = (n_e^b)^{-1} \sum_j \mathbf{1}[b_{ej} \le 0.5]$; the raster convention is zero for shade and one for sun. Except for hybrid azimuth, each method uses this value when present and the surrogate s_e^{az} otherwise. The hybrid-azimuth method uses s_e^{az} everywhere.

The 26°C reference is the conventional UTCI boundary between no thermal stress and moderate heat stress [12]. It supplies an engineering reference rather than a personalized safety threshold. The main run uses a 5 m maximum interval on nominal 1 m rasters. Section 4.4 compares routes selected at coarser spacings on this same reporting field.

Reported route quantities are length-weighted edge summaries:

$$H(P) = \frac{\sum_e l_e h_e}{L(P)}, \qquad \bar{U}(P) = \frac{\sum_e l_e \bar{u}_e}{L(P)}, \tag{5}$$

$$S(P) = \frac{\sum_e l_e s_e}{L(P)}, \qquad U^{\max}(P) = \max_e u_e^{\max}. \tag{6}$$

H is mean positive UTCI exceedance, expressed in kelvins; $\bar{U}$ and $U^{\max}$ are UTCI values in degrees Celsius. S is a shade fraction. Because sampling restarts at segment boundaries, these quantities are numerical proxies rather than uniform line integrals. As a post hoc check of normalization, we also calculate $J(P) = L(P)H(P)$, an exceedance–distance index in K m. It tests whether a lower route average offsets added distance. Neither H nor J is a validated physiological dose.

3.3 Objectives and method variants

We define the composite cost as an engineering score. It combines distance and time with thermal, scenic, traffic, and slope penalties; Appendix C specifies the terms and normalizations. All are nonnegative. The thermal term is

$$\phi_e = \frac{[\bar{u}_e - \tau]_+}{8\,\mathrm{K}} + \frac{[u_e^{\max} - \tau]_+}{10\,\mathrm{K}} + \frac{h_e}{6\,\mathrm{K}} + 0.45(1 - s_e), \tag{7}$$

where $[x]_+ = \max(0, x)$. Its correlated UTCI and shade components can count related radiative information more than once. Neither its coefficients nor the contextual proxies are fitted to route-choice or physiological observations.

Table 1 defines the four objective variants of this implementation. Distance and time remain active in all four. The focused methods are therefore not pure single-feature optimizations. For reporting, selected routes are evaluated with full features; the hybrid shade outcome remains a surrogate and is not comparable with direct shade.

3.4 Capped search and fallback

The solver first computes an unconstrained minimum-cost path and returns it if it meets D_0. Otherwise it explores cost–distance labels in increasing cost order. At a node, a label dominates another if it has no greater length and no greater cost. Labels are pruned when accumulated length plus shortest remaining distance exceeds D_0. Paths may not repeat nodes.

Table 1: Configured objective variants. A_e is the distance–time term in Appendix C; all environmental penalties also use its edge-length scale.

Method	Selection objective
Shortest	Dijkstra search minimizing $L(P)$.
Hybrid azimuth	A_e plus twice the thermal term, using s_e^{az} on all edges.
Direct shadow	A_e plus twice the shade-only part of ϕ_e.
UTCI exceedance	A_e plus twice the $h_e/(6\,\mathrm{K})$ part of ϕ_e.
Composite	Full thermal, scenic, traffic, and slope cost.

The dominance registry retains at most eight labels per node, ordered by cost then length. A newly queued label can survive even when omitted from that registry, so this does not bound the queue size. The label loop stops after 20,000 queue removals or 1.5 s, with timing beginning after the initial cost-path calculation. A capped search returns a feasible length-shortest fallback when available. If feature removal leaves no path within the original ceiling, the request fails. The final acceptance check enforces $L(P) \le D_0$ within 10^{-6} m numerical tolerance.

We record whether a returned route came from the unconstrained cost path, label search, or shortest-path fallback, along with cap termination and failure type. The caps preclude a general optimality guarantee and do not bound end-to-end latency. Appendix A compares the search with exact enumeration on small graphs.

The implementation propagates the original-network ceiling, chooses shortest parallel edges for fallback, preserves sampled exceedance through objective projection, and uses above-ground building heights. These corrections matter numerically: $\mathbb{E}[\max(0, U - \tau)]$ cannot be replaced with $\max(0, \mathbb{E}[U] - \tau)$. The earlier execution archive is retained separately.

4 Data and Analysis

4.1 Study domain and inputs

The study uses one modeled Washington, DC snapshot labeled June 19, 2026, 14:00. The rerun explicitly uses UTC−04:00 for the hybrid solar calculation. Sustainable DC 2.0 reports a 13.3% walking commute share in 2016 [11]; the sampled OD pairs nevertheless represent covered network nodes, not observed trips.

Table 2 lists the input sources. SOLWEIG-GPU [15] generated the nominal 1 m UTCI and shadow fields from urban-surface inputs and hourly Open-Meteo forcing [27]. These fields are held fixed and have no independent validation here. The versioned graph and rasters are stored with SHA-256 checksums; all 500 archived shortest distances reproduce exactly. The original graph has 61,543 nodes and 174,718 directed edges. For the hybrid proxy, building height is $\max(0, \mathrm{DSM} - \mathrm{DEM})$ on the aligned source grids; absolute surface elevation must not be interpreted as height above ground. Frozen routing inputs enable rerouting, but the retained provenance still lacks the forecast retrieval time, selected forecast model, and environmental-model release needed to regenerate the thermal field exactly.

Table 2: Input layers. Different source years and street-centerline sampling limit spatial interpretation.

Layer	Source and role
Walking network	OpenStreetMap via Geofabrik/OSMnx; directed centerlines, not verified sidewalks [2, 21].
UTCI and shadow	SOLWEIG-GPU modeled snapshot; thermal and direct-shade features [15].
Terrain/buildings	NOAA-hosted DC OCTO 2022 LiDAR and DEM; grade and height proxies [19, 20].
Canopy/land cover	LiDAR canopy heights and DC 2020 layers; greenery, water, and surrogate shade [6, 7].
Traffic	DDOT 2023 AADT lines; normalized traffic proxy, not sound measurements [9].

4.2 OD sample and experimental design

The original generator uses the largest weakly connected graph component, then retains endpoints with finite UTCI and shadow raster values. With seed 42, it samples 500 distinct ordered node pairs whose shortest network distances lie between 400 and 3,000 m. We reuse those exact pairs. Their shortest distances range from 412.9 to 2,998.9 m. Figure 1 locates the endpoints and shows the network-distance distribution used for eligibility. Large parts of the District have no sampled endpoints, so these locations do not establish district-wide coverage. Both maps use EPSG:32618 and frozen DDOT/DC GIS roads and water for geographic context; those display layers do not enter routing or exposure calculations [8, 10].

Each pair is evaluated with five methods at four allowances, giving 10,000 attempts. The 2,000 shortest-distance records repeat 500 paths; the analysis checks their edge sequences across allowances. Environmental routes can change with allowance-specific corridors, features, and search. Records from the same OD pair are not independent observations. The 5 m run uses 32 batch shards, with fixed method order and separate caches for direct-shadow and hybrid features. Manifests record each configuration and source-file hashes.

4.3 Matched comparison and uncertainty

Let $P_{i\delta}^{0}$ be the matched shortest route and $P_{i\delta}^{m}$ the route returned by method m. Positive improvements are

$$\Delta H_{i\delta m} = H(P_{i\delta}^{0}) - H(P_{i\delta}^{m}), \tag{8}$$

with analogous $\Delta\bar{U}$; shade change reverses the order. Extra distance is $100(L(P_{i\delta}^{m})/L_{0}^{*} - 1)$.

At each allowance, the primary comparison uses the intersection $\mathcal{C}_{\delta}$ of successful records for all five methods. Matching includes study, configuration hash, run identifier, raster timestamp, OD identifier, and allowance; duplicates are rejected. Shards retain their original identifiers, and matching stays within each shard. The common set contains 2,000 of 2,000 pair–budget cases from 500 OD pairs. Pooled means weight these cases equally and include shortest-path fallbacks. We also count records with lower H but higher J, using 10^{-9} K and 10^{-6} K m tolerances to exclude rounding noise.

For a paired method contrast, we compute the mean difference in recorded H on the same common records. Percentile intervals use 2,000 bootstrap resamples of whole OD pairs, retaining each pair's available allowances, with seed 42. These exploratory 95% intervals are not adjusted for multiple comparisons. They summarize variability within this one modeled snapshot and do not account for microclimate error or selection of the successful subset.

4.4 Sampling and coefficient checks

We rerun all 500 OD pairs, methods, and allowances at 5 m, holding the original graph, rasters, threshold, weights, and search caps fixed. The earlier 20 m experiment supplies a second complete set of selected paths. We rescore all 10,000 of those paths at 5 m, then compare the two sets on the same reporting field. Matching across grids uses OD, allowance, and method, after verifying identical input checksums and unchanged routing code. The original shortest paths must have identical edge sequences and common-field scores in both runs.

On the first 100 serialized OD pairs at a 20% allowance, we test 16 composite settings: the 5 m reference; removal and doubling of each thermal, shade, scenic, traffic, and slope coefficient; thresholds of 24 and 28°C; and intervals of 10, 20, and 40 m. Each coefficient changes alone; distance and time weights stay fixed. Every selected path is scored at 5 m and $\tau = 26$°C. Path changes use ordered edge sequences. For coarser sampling, we also compare each selected path's own-grid score with its 5 m score. The 5 m field is a numerical reference, not ground truth or proof of convergence at 1 m. Failures remain in the denominator. A repeated reference setting checks for path changes under the caps.

5 Results

5.1 Feasibility and route sources

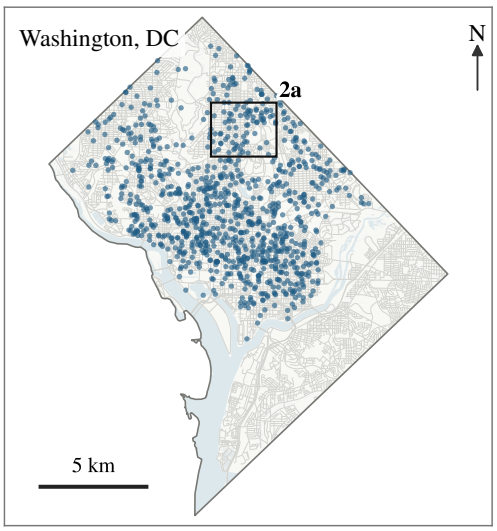
The corrected run returns 10,000 routes from 10,000 attempts, all within their original-network distance ceilings. Environmental fallback accounts for 4.90–13.00% of attempts across the four settings. These are feasible walking routes; fallback does not establish that a cooler route was found. The common comparison includes every such fallback. Recorded end-to-end medians range from 0.205–3.079 s, with 90th percentiles of 0.771–8.059 s and a maximum of 18.728 s. These measurements include corridor and feature extraction; shared caches, fixed method order, and host differences preclude a controlled speed ranking.

5.2 Exposure and added distance

Table 3 compares the same 2,000 OD–budget cases for every method. Mean exceedance reductions range from 0.91–0.94 K, with 4.59–6.34% extra distance. The shortest-route mean is 4.609 K. The composite has 0.002 K lower mean H than the UTCI-focused setting (OD-cluster interval [−0.020, 0.026] K). The interval includes zero. The UTCI-focused setting has 0.027 K lower mean H than the direct-shadow setting (interval [0.011, 0.044] K).

Figure 2 links one recorded route comparison with all 500 common pairs at a 20% allowance. The mapped example adds 8.3% distance and lowers H by 1.04 K. None has higher H than its matched shortest route. Selection minimizes a sum of distance, time, and

(a) Sampled endpoints



Blue: sampled endpoints; box: Fig. 2a map extent

(b) Network-distance eligibility

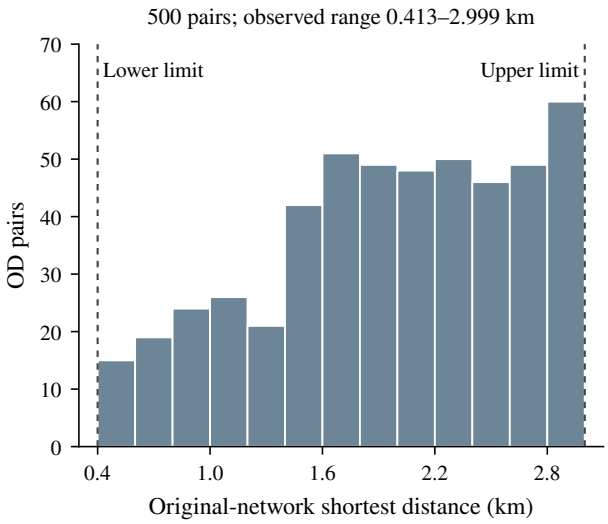


Figure 1: Sample and distance eligibility. (a) Endpoint records for 500 OD pairs within the District; repeated endpoints can overlap. The box locates Figure 2a. (b) Original-network shortest distances in 200 m bins, bounded by the 400–3,000 m sampling limits. Distances are network lengths, not straight-line separation.

Table 3: Matched outcomes at 5 m sampling on 2,000 OD–budget cases. Positive ΔH and ΔJ favor the environmental route; ΔS is added shade. Intervals resample whole OD pairs. The hybrid's shade comparison is omitted because it uses a different surrogate. Fallbacks remain in the outcomes; their percentages use all 2,000 attempts per method.

Method	Mean ΔH (K) [95% interval]	Mean ΔJ (K km)	Mean ΔS	Extra (%)	Fallback (%)
Shortest	0.000	0.000	0.000	0.00	0.0
Hybrid azimuth	0.942 [0.891, 0.993]	1.529	—	5.21	7.7
Direct shadow	0.910 [0.862, 0.961]	1.372	0.135	6.34	13.0
UTCI exceedance	0.937 [0.888, 0.988]	1.435	0.131	6.29	12.8
Composite	0.939 [0.887, 0.991]	1.572	0.130	4.59	4.9

length-scaled thermal costs; it does not directly minimize the normalized H.

Lower mean exceedance can accompany a higher exceedance–distance index. Direct shadow produces 16 such records (0.8%). The composite produces 4, and the UTCI setting produces 0. Direct-shadow routing also raises H in 4 cases. Table 3 reports both H and J. Hybrid shade uses a surrogate, so its ΔS is omitted; the raw records also report how much direct-shadow routing relies on surrogate fallback.

For the composite, mean ΔH changes from 0.698 K at a 5% allowance to 1.098 K at 30%, with mean extra distance of 2.40% and 6.17%, respectively. These summaries use the same matched OD pairs at each allowance. A larger ceiling expands the feasible path set on a fixed graph. The implementation also rebuilds its corridor and applies caps, so the returned route need not improve monotonically for an individual pair.

5.3 Sampling and coefficient sensitivity

All 1,600 sensitivity attempts succeed, and every selected path has 5 m rescoring. Table 5 shows the one-factor weight changes. Removing or doubling a coefficient changes 12–96 of 100 common paths, depending on the coefficient. All exposure contrasts in that table use $\tau = 26$°C and 5 m samples, so the numbers reflect selected paths rather than changed reporting definitions.

Changing the selection threshold to 24 and 28°C changes 42 and 35 paths, respectively. On the common 5 m reporting field, mean H is 0.117 K higher than the default at 24°C and 0.017 K lower at 28°C. These results quantify sensitivity to two nearby references; they do not calibrate a personal heat-stress threshold.

Table 4 compares the complete 20 m and 5 m cohorts on the common 5 m field. The composite changes 70.3% of its paths. The 100-pair check separately finds a median same-path error of 0.099

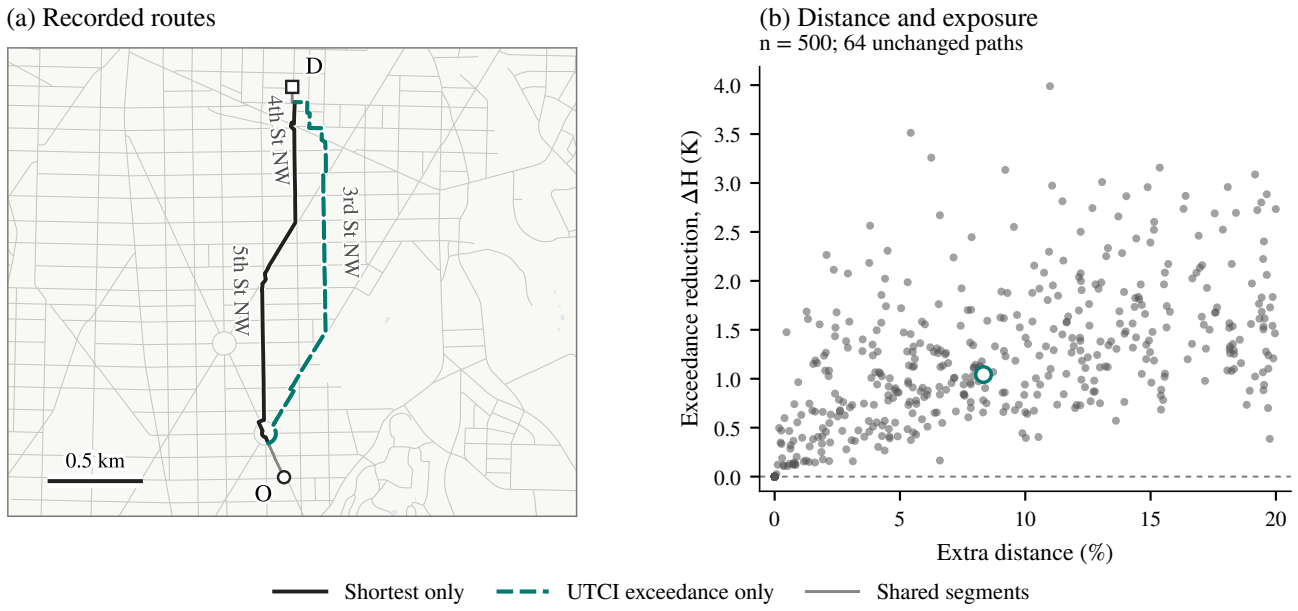


Figure 2: Route change and paired outcomes at 5 m sampling and a 20% distance allowance. (a) Recorded paths for `od_00133`, nearest the median paired exceedance reduction. O/D denote origin/destination; gray segments are shared. (b) Every point is a common OD pair. Positive $\Delta H = H_{\text{shortest}} - H_{\text{UTCI}}$ means lower mean positive exceedance above 26°C. The outlined point identifies (a). Map coordinates use EPSG:32618.

Table 4: Full-cohort sampling comparison on 2,000 common OD–budget cases. Both sets of routes are scored at 5 m. The final column is $H(P_{20}) - H(P_5)$; positive values favor 5 m selection. Intervals resample whole OD pairs.

Method	Changed paths (%)	ΔH, 20 m routes (K)	ΔH, 5 m routes (K)	5 m gain (K) [95% interval]
Hybrid azimuth	70.20	0.882	0.942	0.060 [0.044, 0.077]
Direct shadow	76.15	0.812	0.910	0.099 [0.081, 0.117]
UTCI exceedance	70.35	0.865	0.937	0.072 [0.052, 0.094]
Composite	70.30	0.877	0.939	0.063 [0.048, 0.077]

Table 5: Composite sensitivity on 100 paired paths at 5 m and 26°C. Entries show removed/doubled coefficients. Positive ΔH means lower exceedance than the default.

Coefficient	Changed paths	ΔH (K)
Thermal	96/30	−0.572/0.032
Shade	23/14	−0.048/0.022
Scenic	35/34	0.000/ − 0.024
Traffic	23/21	0.019/ − 0.038
Slope	12/14	0.010/ − 0.021

K at 20 m, with a 90th percentile of 0.252 K. The repeated 5 m reference changes 0 paths. These checks distinguish changes in the selected routes from changes in their reported exposure.

6 Discussion and Limitations

The experiment compares engineering objectives within one search implementation. The fixed cohort and explicit fallback records support paired comparisons of those objectives; independent routing algorithms remain untested. The exact checks expose cap-induced losses even on 33-node networks (Appendix A); no optimality gap is available for the DC requests. A city-scale comparison with an exact or tightly bounded constrained-path solver remains necessary before claiming search quality.

The spatial and temporal scope is one modeled DC hour. The study cannot establish transfer across cities, seasons, neighborhoods, or times of day, and the fixed field ignores changes during a walk. The OD sample targets covered network nodes rather than observed demand. It therefore cannot estimate population benefits or typical pedestrian trips.

Street centerlines may lie in sun while a sidewalk is shaded, or the reverse. Source layers have different vintages; the traffic join uses a nearest-midpoint proxy in geographic coordinates without a maximum distance. Canopy does not guarantee shade at the pedestrian's position. Hybrid shade mixes geometry with cool UTCI samples and lacks independent shadow validation.

The composite remains uncalibrated. Thermal terms overlap, scenic and traffic scores are proxies, and the uncertainty input is zero. The one-factor checks describe local sensitivity at one allowance; they do not identify optimal weights or cover interactions. The common-field comparison tests sensitivity to 20 m versus 5 m routing. It does not establish convergence at the raster-cell scale or physical accuracy.

The study has no field transects, participant study, or health measurements. Mean exceedance and exceedance–distance can disagree. Independent measurements and sidewalk-level networks are needed to test the modeled differences at pedestrian height and position before they support route advice.

Code and Data Availability

The code and experiment files includes code, the exact OD pairs, compressed route records, source/input checksums, analysis scripts, and figure inputs. Git LFS stores the versioned graph and rasters. The README gives rerun and analysis commands. Manifests preserve each shard's configuration and the routing source hashes; the original execution environment and thermal-field generation are only partly documented. The archived run and joint-weight exploration are retained separately and do not enter the revised result tables.

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A Exact Search Check

We generate 100 graphs in each of three families with seed 42: 17-node layered directed graphs (five layers of width three), nine-node cyclic directed graphs, and 11-node chains with two parallel edges per step. Edge lengths are 40–180 m; thermal and contextual features are generated numerical values. An independent depth-first enumerator retains every simple edge path, including parallel-edge choices. It shares the configured edge-cost function with the router, so this tests search rather than validating the cost model.

Three hundred graphs, four allowances, and UTCI/composite objectives give 2,400 reference cases. Of these, 1,395 have a constrained optimum strictly above the unconstrained optimum. With

the wall-clock cap disabled, the default eight-label registry and 20,000-expansion limit match every reference within relative tolerance 10^{-8} and respect every distance ceiling. Removing the expansion cap with registry sizes one, eight, or 32 also matches all cases. A deliberately tight 20-expansion condition matches 1,687 of 2,400 cases; its capped fallbacks can be suboptimal. Frozen generated graphs, raw results, and the enumerator are shared. Agreement on these small graphs supplies no optimality bound for the DC network.

To activate the actual default cap, we also generate 25 chains with 32 binary parallel-edge choices: 33 nodes and 2^{32} possible paths each. Short edges have integer lengths of 40–70 m and exceedance of 8–12 K; longer edges add 10–50 m with exceedance of 0–4 K. An independent dynamic program retains minimum cost at every integer distance. Across 200 graph–objective–allowance cases, the default count limits match 50 exact references and truncate 150 searches. Mean relative cost gap is 42.3%, with a maximum of 136.0%. These deliberately difficult cases diagnose cap behavior; they do not estimate error on the DC network. Both checks disable the wall-clock cap.

B Engineering Feature Definitions

Let $\mathrm{clip}(x) = \min(1, \max(0, x))$. Canopy and land cover are sampled on the densified centerline; building height is additionally sampled at lateral offsets of one-half and one estimated street width.

Let c_e be the fraction of canopy samples at least 2 m high; k_e the mean canopy height divided by 12 m and clipped; v_e the larger of canopy- and land-cover-derived greenery fractions; and w_e the water fraction. The scenic proxy is

$$g_e = \mathrm{clip}(0.60c_e + 0.25k_e + 0.10v_e + 0.05w_e). \tag{9}$$

For nearest-midpoint AADT a_e and the traffic layer's maximum $a_{\max}$, the traffic proxy is $n_e = \log_{10}(a_e + 1)/\log_{10}(a_{\max} + 1)$. This is not a decibel scale.

For solar altitude α, azimuth θ, bearing β_e, and estimated street width W_e, define

$$q_e(z) = \mathrm{clip}\left(\frac{z|\sin(\theta - \beta_e)|}{W_e \max(\tan\alpha, 10^{-6})}\right). \tag{10}$$

If explicit width is missing, $W_e = \max(4, 3.2p_e)$ m for p_e lanes, or 10 m if lane count is also missing. With mean and maximum canopy/building heights, use $z_e^c = \max(\bar{z}_e^c, 0.5z_e^{c,\max})$ and $z_e^b = \max(\bar{z}_e^b, 0.6z_e^{b,\max})$. The geometry shade proxy is

$$s_e^{\mathrm{geom}} = \mathrm{clip}\Big(\max\{c_e, q_e(z_e^b), c_e q_e(z_e^c)\}\Big), \tag{11}$$

set to one below the horizon. For local fractional hour t,

$$\tau_c(t) = [25.5 + 3.2\cos(2\pi(t - 14)/24)]^\circ\mathrm{C}, \tag{12}$$

$$s_e^{\mathrm{cool}} = (n_e^u)^{-1} \sum_i \mathbf{1}[u_{ei} \le \tau_c(t)], \tag{13}$$

$$s_e^{\mathrm{az}} = \max(s_e^{\mathrm{cool}}, s_e^{\mathrm{geom}}). \tag{14}$$

At $t = 14$, the cool threshold is 28.7°C. These coefficients define a hybrid surrogate, not a fitted shadow model.

C Composite Cost

Let signed elevation change be Δz_e, grade $m_e = \Delta z_e/l_e$, and $r_e = 10|m_e| + [\Delta z_e]_+/(25\ \mathrm{m})$. Edge time is t_e, estimated as $l_e/(1.35\ \mathrm{m\,s^{-1}})$ when unavailable. With ϕ_e from Eq. 7, the default cost is

$$A_e = 0.15\frac{l_e}{1000\ \mathrm{m}} + 0.20\frac{t_e}{600\ \mathrm{s}}, \tag{15}$$

$$C_e = A_e + \frac{\max(l_e, 1\ \mathrm{m})}{100\ \mathrm{m}}\big[\phi_e + 0.45(1 - g_e)^2 + 0.45n_e^2 + 0.35r_e + 0.15\sigma_e\big], \tag{16}$$

$$C(P) = \sum_{e \in P} C_e. \tag{17}$$

All ratios are dimensionless. $\sigma_e = 0$ throughout this experiment. The one-factor checks change the corresponding coefficients and shade multiplier separately. The 1 m floor prevents a zero environmental scale on a malformed zero-length edge. These formulas specify the engineering score used in the corrected rerun.